# FEDERAL SPECIFICATION

# LEATHER, CATTLEHIDE, HARNESS, VEGETABLE-TANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for vegetable tanned cattlehide harness leather (see 6.1).

1.2 Classification. The leather shall be furnished in the following types and classes, as specified (see 6.2):

Type I - Sides.
Type II - Backs.
Class 1 - Mildew-resistant treated.
Class 2 - Not mildew-resistant treated.

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

KK-L-311 - Leather; Methods of Sampling and Testing.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-43432 - Leather, Packaging and Packing of.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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3. REQUIREMENTS

3.1 Material. The leather shall be vegetable tanned, full grain or corrected grain cattlehide. Corrected grain leather shall be full grain leather that has been lightly snuffed on the grain side to remove superficial blemishes. The flesh side shall be smooth and free of loose flesh and may be shaved or buffed to provide greater uniformity of thickness. Use of bull hides is prohibited (see 6.3).

3.2 Physical requirements.

3.2.1 Trim (form and area). The leather shall be furnished in the form of sides or backs as required for the type specified (see 6.2) and shall be trimmed to conform with the definition of sides or backs as listed under "Definition of Terms" in KK-L-311. The area in terms of square feet shall be legibly marked in the butt area on the flesh side of the leather.

3.2.2 Color. The color of the leather shall be as specified (see 6.2).

3.2.3 Finish. Unless otherwise specified (see 6.2), the leather shall have a light application of finish on the grain side.

3.2.4 Weight and thickness. The leather shall be of the weight and thickness specified (see 6.2), and as shown in table I.

TABLE I. Weight and thickness

| Weight | Thickness (ounces) 1/ |
|---|---|
| Extra light | 8 to 9-3/4 |
| Light | 10 to 11-3/4 |
| Medium | 12 to 13-3/4 |
| Heavy | 14 to 15-3/4 |
| Extra Heavy | 16 and above |

1/ 1 ounce = 1/64 inch.

3.2.5 Breaking force. Not less than 80 percent of the specimens tested shall meet the minimum breaking force values specified in table II, column 1, and no specimen shall have a breaking force value below that shown in table II, column 2, when tested as specified in 4.3.

TABLE II. Breaking force

| Thickness (ounces) | Column 1<br>Breaking force minimum (pounds) | Column 2<br>Breaking force minimum (pounds) |
|---|---|---|
| 8 to 9-3/4 | 135.0 | 110.0 |
| 10 to 11-3/4 | 170.0 | 140.0 |
| 12 to 13-3/4 | 210.0 | 170.0 |
| 14 to 15-3/4 | 240.0 | 200.0 |
| 16 and above | 260.0 | 230.0 |

3.2.6 Elongation. Not less than 80 percent of the specimens tested shall have elongation values of 20 percent or less under the loads shown in table III when tested as specified in 4.3.

TABLE III. Elongation load

| Thickness (ounces) | Load (pounds) |
|---|---|
| 8 to 9-3/4 | 110.0 |
| 10 to 11-3/4 | 140.0 |
| 12 to 13-3/4 | 170.0 |
| 14 to 15-3/4 | 200.0 |
| 16 and above | 230.0 |



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3.2.7 Cracking. The leather shall not crack when tested as specified in 4.3.

3.3 Chemical requirements.

3.3.1 The leather shall conform to the chemical requirements of table IV when tested as specified in 4.3.

TABLE IV. Chemical requirements

| Characteristic | Minimum | Maximum |
|---|---|---|
| Chloroform-soluble material, percent 1/ | 15.0 | 35.0 |
| Water-soluble material, percent 1/ | -- | 18.0 |
| Total ash, percent 1/ | -- | 5.0 |
| Degree of tannage | 50.0 | 75.0 |
| Acidity (pH) | 3.3 | 4.5 |

1/ Calculated on the moisture free basis.

3.3.2 Class 1. When class 1 is specified (see 6.2), the leather shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.3.

3.4 Workmanship. The finished leather shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with KK-L-311 except when otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. The classification of defects found during the examination shall be in accordance with the lists shown in 4.2.2.1 and 4.2.2.2. The sample unit for these examinations shall be one side or one back. The lot size shall be expressed in units or one side or one back. The sample size for thickness determination shall be 15, regardless of lot size and the rejection number shall be 2.



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4.2.2.1 Examination for visual characteristics.

| Examine | Defect |
|---|---|
| Trim | Not as specified. |
| Color | Not as specified |
| Finish | Unless otherwise specified, grain side does not have application of finish. |
| Quality of leather | Grain not as specified.<br>Flesh side not smooth or contains areas of loose flesh in excess or four square inches in total area.<br>Stain, burn mark, cut, scratch, hole, brand, hip mark, rough shoulder, heavy vein, deep fat wrinkle, boney or flanky area, or other surface imperfections which impair the cutting value of more than 12 percent of the total area of the leather. |

NOTE: Light, well healed, scratches or grub holes; light fat wrinkles; or slight stains shall not be classified as defects.

4.2.2.2 Examination for dimensional characteristics.

| Examine | Defect |
|---|---|
| Area (expressed in terms of square feet) | Less than area marked on leather. |
| Marking | Omitted. |
| Thickness 1/ | Not as specified. |

1/ The thickness shall be determined in accordance with method 1011 of KK-L-311.

4.2.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of MIL-L-43432 to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with.

4.3 End item testing. The methods of testing specified in KK-L-311, wherever applicable and as listed in table V, shall be followed. Sampling procedures shall be in accordance with KK-L-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exists:

a. Any test failure occurs for breaking force.
b. More than three test failures occur for elongation.
c. More than one test failure occurs for cracking.
d. Any composite value fails to meet the specified requirement.



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TABLE V. Tests

| Characteristic | Requirement paragraph | Test method | Requirements applicable to Sample unit | Composite |
|---|---|---|---|---|
| Material identification | 3.1 | 1/ | -- | -- |
| Breaking force | 3.2.5 | 2021 | X | -- |
| Elongation | 3.2.6 | 2021 | X | -- |
| Cracking | 3.2.7 | 4.3.1 | X | -- |
| Chloroform-soluble material | 3.3.1 | 6341 | -- | X |
| Water-soluble material | 3.3.1 | 6341 | -- | X |
| Total ash | 3.3.1 | 6421 | -- | X |
| Degree of tannage | 3.3.1 | 6631 | -- | X |
| Acidity (pH) | 3.3.1 | 6621 | -- | X |
| Paranitrophenol | 3.3.2 | 6711 | -- | X |

1/ The supplier shall submit a certificate of compliance for this characteristic.

4.3.1 Cracking. The leather shall be tested for cracking in accordance with method 4011 of KK-L-311. The diameters of the mandrels used for testing the leather shall be as follows:

| Thickness of leather (ounces) | Diameter of mandrel (inches) |
|---|---|
| 8 to 9-3/4 | 5/16 |
| 10 to 11-3/4 | 3/8 |
| 12 to 13-3/4 | 7/16 |
| 14 to 15-3/4 | 1/2 |
| 16 and above | 9/16 |

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Leather shall be packaged in accordance with the applicable requirements of MIL-L-43432.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Leather shall be packed in accordance with the applicable requirements of MIL-L-43432.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-L-43432.

6. NOTES

6.1 Intended use. The leather is primarily intended for use in safety belts and harness straps.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2 and 3.2.1).



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(c) Color required (see 3.2.2).
(d) When finish is not required (see 3.2.3).
(e) Weight and thickness required (see 3.2.4).
(f) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Bull hides. Bull hides are excluded from the cattlehide to be used for harness since they do not produce good-quality leather. A bull hide usually ranges in weight from 75 to more than 100 pounds in the green, salted state. It is characterized by very thick and rough neck and shoulders.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
Navy - SA, MC, SH
Air Force - 82, 85

User activity:

Army - EL

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

AGR
COM-NBS
GSA-FSS
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# SPECIFICATION ANALYSIS SHEET

Form Approved
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**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Leather, Cattlehide, Harness, Vegetable Tanned; KK-L-156c

| ORGANIZATION | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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